Mathieu Audibert

Data Engineer

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Education

ESILV September 2025 – August 2027
MSc Data Engineering & AI *Nanterre, Hauts-de-Seine*

- **Relevant Coursework:** Advanced OOP (Java, Spring Boot), ETL & Pipeline Orchestration (Kafka, Airflow), DevOps (Ansible, Terraform, AWS, K8s, Docker), Databases (Oracle), Machine Learning & Clustering (Scikit-learn, TensorFlow).

Efrei Panthéon-Assas September 2022 – August 2025
BSc Developpement & Data *Villejuif, Val-de-Marne*

- **Relevant Coursework:** OOP (Java, Spring Boot), Web Development (JavaScript, PHP, APIs), Cybersecurity (OWASP, CVEs, UML), Databases (PostgreSQL, MySQL, MongoDB), Linux Systems & Kernel.

Lycée Alexandre Dumas September 2019 – May 2022
Baccalauréat NSI, Physics-Chemistry & Complementary maths *St-Cloud, Hauts-de-Seine*

- **Relevant Coursework:** Graph Theory, Data Structures, Networking, Algorithms, Thermodynamics, Waves & Signals, Electricity.

Experience

Crédit Agricole Assurances September 2025 – August 2027
Data Engineer · Apprentice *Paris XV, Ile-de-France*

- Featured new management rules on Java Spring Boot Kafka Streams & ksqlDB Big Data pipelines.
- Upgraded an API handling insurance claims from Spring Boot 3 to Spring Boot 4 (Spring Security, K8s, SVC, Composites), enhancing performance and security.
- Refactored 8+ projects from legacy SQL to clean Spark sessions, with Jenkins pipelines, anti-pattern cleanup (double-brace initialization, God classes) and code coverage up to ≥ 75%.
- Updated technical documentation and coached colleagues on Git workflows and code quality improvements.
- Deployed and assisted deployments (SAFe context) on all environments using Artifactory, Docker, K8s and ArgoCD.

Caisse Primaire d'Assurance Maladie September 2024 – July 2025
Concepteur Développeur d'Applications · Apprentice *Créteil, Val-de-Marne*

- Rebuilt a local IT asset management tool in PHP, used by secretaries and technicians to streamline administrative processes.
- Designed and developed a Symfony web platform supporting the monitoring of 800k beneficiaries of the MisAS health aid campaigns (front-end and back-end).
- Introduced a DevOps culture to bring the department up to today's standards.

Société Générale April 2024 – June 2024
DevOps Hybrid Cloud · Trainee *Val-de-Fontenay, Val-de-Marne*

- Discovered hybrid-cloud network and architecture, DevOps practices and agility.
- Built a Python script to interrogate and report AWS Public Cloud transit gateways.

Scientific Paper

Intelligent Monitoring of Micro-Services Prototype | *Anomaly Detection, Dynamic Weight Selection, Log Analysis*
Demo paper addressing the question: *"To what extent can a dynamic model-selection mechanism, guided by a Region of Competence and Dynamic Weight Selection, improve anomaly detection in heterogeneous microservice systems compared with single static detectors?"*

Projects

Prodigieux Corp. | *Self-hosted Cloud, Python, Ubuntu Server, Server administration*

- Self-hosted cloud on a repurposed Dell Latitude (serving as a server) and a Raspberry Pi.
- Sentra: API to remotely administrate the server (users/groups, files, power on/off) through SSH tunnels and a SSH client.
- Dived in networking by configuring Wake-on-LAN with broadcast forwarding.

- WireGuard VPN on the Raspberry Pi to bounce to the server's local network.

Gideon | *Python, Kafka, Airflow, Grafana, PostgreSQL, PostGIS*

- Military topology interactive map built on two large GeoJSON datasets ("Ukraine roads" & "Ukraine Points of Interests").
- ETL + ELT streaming pipeline ingesting raw data into PostGIS, building SQL analytical tables, publishing replayed events to Kafka and serving a Grafana dashboard.

Local Machine Learning Flow | *Python, Cloud Computing, MLOps, CI/CD*

- MLOps ecosystem automating the machine learning lifecycle through local cloud computing (FastAPI, LocalStack AWS emulation, CI/CD).
- Orchestrated data retrieval, model training and evaluation through serverless functions in a fully containerized environment.

Eco-pulse | *Python, Cloud Computing, Airflow, CI/CD*

- ELT pipeline ingesting global carbon emissions: extraction into B2 buckets, cleaning/transformation and loading into Google Cloud BigQuery.
- Centralized secrets management with Aiven Valkey (environment variables, API keys), WAF and partial SSRF / third-party protections.

Cooloc | *Python, React, Docker, PostgreSQL, UML, SonarQube, CI/CD, OWASP*

- Shared-housing management web platform: full design and implementation (database, patterns, backend, frontend, security, deployment).
- RESTful API on a custom Python HTTP server (no frameworks allowed), React frontend, complete UML documentation.
- OWASP-compliant security: RBAC, ANSSI-grade passwords, SQL/XSS prevention, CSRF & JWT tokens, GDPR compliance.

Challenges

Crédit Agricole Data x AI Hackathon | *September 2026*
Uniting 160+ participants around leveraging AI to enhance insurance processes: I placed 2nd by building agents that modified fonctionnal documentation based off developpers tasks states. It fetched the Jira Tickets from the developpers, created 2 dashboards for managers and wrote propositons of documentation improvements. Pitched it in front of the whole audience.

Solved a Jane Street puzzle | *August 2026*
"Andy's Afternoon Amble", a probability puzzle solved with basic Java 21.

Certifications

November 2025 • AWS Cloud Quest Practitioner - Training badge | *ID: 6584e3b1-db1e-49a4-adfe-4e0b82eaf6fd*

April 2025 • Concepteur Developpeur d'Applications | *ID: 988224272263*

Skills

Programming languages: Java, Python, Rust (learning), PHP, SQL, Bash, Kotlin (learning)
Technologies: AWS, GCP, Kafka, Oracle, PostgreSQL, MongoDB, GraphQL, Docker, Kubernetes, Git, GitHub/GitLab CI, ArgoCD, Artifactory
Concepts: Linux Kernel, Cryptography, AI & Neural Networks, APIs, DevOps, Clean Code, Cloud Computing, OOP
Frameworks: Spring Boot, FastAPI, Jupyter, Polars, Symfony
AI tools: Copilot, OpenCode (Amazon Nova, Kimi K3, Mistral 7b)
Languages: French (native), English (fluent), Spanish (partial), Portuguese (notions)
Soft skills: Autonomous, Communicative, Punctual, Curious

Interests

Sports: Football, Volleyball, Motorsports
Art: Paintings, Cinema, Music
Technology: Medical assistance through robotics, Information powered by Computer Science
Finance: Jane Street Puzzles, ETFs, Monte Carlo, Bayes' Theorem